

Federated IoT *Anomaly Detection*

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STATUS

Active research project, January 2026 to present. Phase 2 (physical Raspberry Pi client) is in progress.

PAPER

On the way: *Federated Autoencoders for Non-IID IoT Anomaly Detection: Protocol Design, Aggregator Comparison, and Stability Analysis*. IEEE conference draft written; multi-seed runs under way before submission.

01 THE PROBLEM

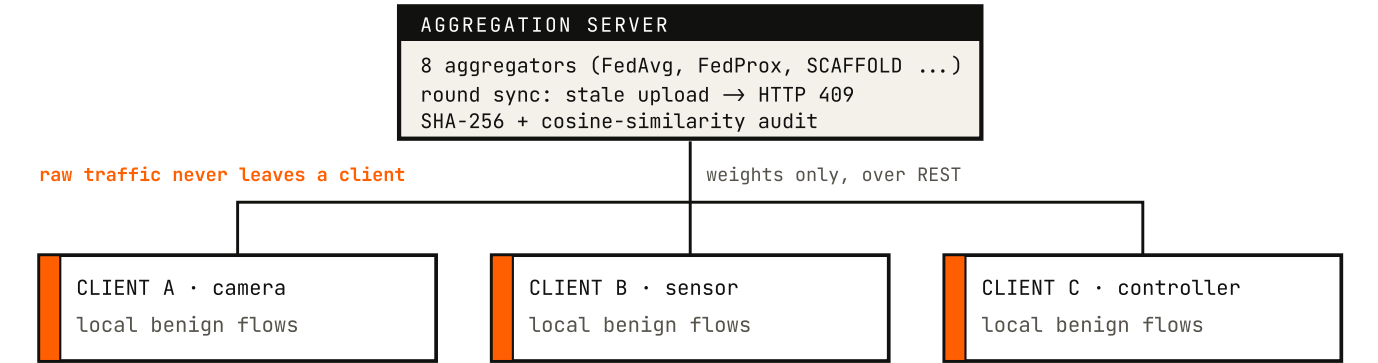
IoT networks are attacked device by device, but the traffic needed to train a detector sits on those devices and is often too sensitive or too heavy to centralise. Federated learning keeps raw traffic local and shares only model weights. The catch is that real devices see very different traffic (non-IID data): a camera sees spoofing, a sensor sees floods, a controller sees probing. We study whether a **federated autoencoder trained on benign traffic only** can still detect all of these attack families, and which aggregation algorithm stays stable when clients disagree.

02 SETUP

CLIENT	DEVICE ROLE	ATTACK FAMILIES HELD OUT FOR EVALUATION
A	Camera	Spoofing
B	Sensor	DoS / DDoS floods
C	Controller	Reconnaissance, web attacks, brute force

Dataset: CIC IoT-DIAD 2024 (Canadian Institute for Cybersecurity), partitioned by device role to create non-IID clients.

03 WHAT WE BUILT

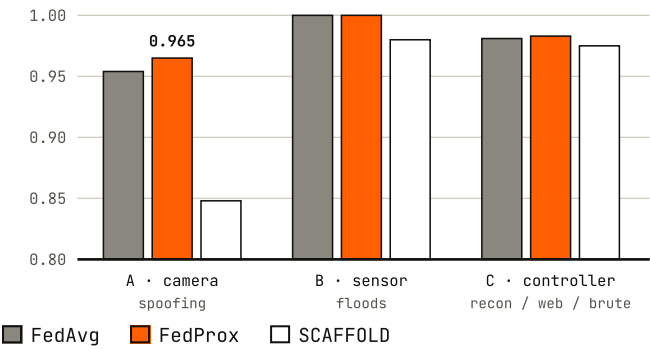


- **Benign-only autoencoder.** A LayerNorm MLP autoencoder (about 45k parameters) learns to reconstruct normal traffic; a flow is flagged when its reconstruction error exceeds the 90th percentile of benign error. No attack labels are needed for training.
- **RFRE identifier encoding.** Network identifiers such as IP addresses are encoded by relative frequency rank instead of being dropped, taking the feature space from 34 to 38 dimensions. MinMax scaling is fitted on benign traffic only.
- **Real federated protocol, not a single-process simulation.** A Dockerised server and three clients talk over REST with round synchronisation (stale uploads are rejected with HTTP 409) and an integrity audit on every update.
- **Eight aggregation algorithms** behind one interface: FedAvg, FedProx, SCAFFOLD, FedAvgM, FedAdam, FedDyn, FedNova and FedRep, with per-round metrics logged for every client.

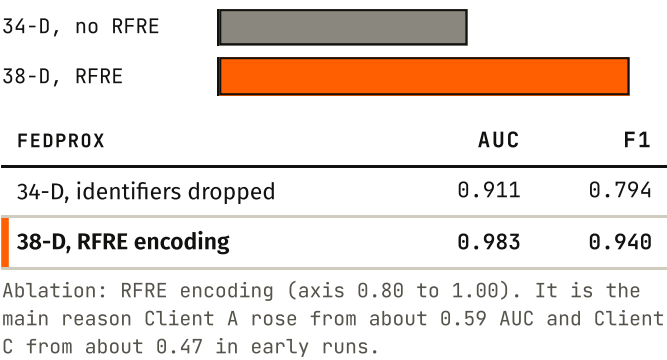
04 RESULTS SO FAR

AGGREGATOR	ROUNDS	LAST AUC	PEAK AUC	LAST F1	LAST LOSS
FedProx ($\mu = 0.01$)	17	0.9825	0.9825 (r17)	0.9404	0.0040
FedAvg	15	0.9784	0.9811 (r5)	0.9396	0.0024
SCAFFOLD	24	0.9343	0.9789 (r1)	0.9132	7.87

Aggregated detection quality on the 38-D RFRE pipeline. Single-run point estimates; multi-seed runs are in progress.



Last-round AUC per client. Spoofing on the camera is the hardest case; floods are near-saturated and should not be over-read.



Ablation: RFRE encoding (axis 0.80 to 1.00). It is the main reason Client A rose from about 0.59 AUC and Client C from about 0.47 in early runs.

05 A NEGATIVE RESULT WE DIAGNOSED

SCAFFOLD matched the others at round 1 (0.979 AUC), then its loss jumped from 0.0062 to 1.20 at round 2 and ended at 7.87. Rather than drop it from the comparison, we traced the cause: with an Adam optimiser on the clients, the control variates SCAFFOLD uses to correct client drift were on the scale of an Adam step, not a gradient.

Measured correction-to-gradient ratio: **22.4×** under Adam versus **0.089×** under SGD. Clamping the correction (the original band-aid) changed nothing. The fix we adopted, SGD with the paper's Option II update and no clamp, tracks FedAvg's loss in a matched smoke test (4.99 vs 4.91).

06 FROM SIMULATION TO A LIVE DETECTOR

The trained FedProx model now runs as a live intrusion detector: packets are captured with Scapy, assembled into flows that mirror the dataset's CICFlowMeter features, and scored in real time, with an attacker console that replays benign traffic and SYN floods. The first live tests did not separate benign from attack traffic. The cause was a domain gap rather than the model: local IPs unseen in training, desktop TCP settings unlike IoT stacks, and a static threshold. We fixed these by injecting deployment IPs into the RFRE table, tuning TCP to IoT-like windows, and calibrating the threshold on a benign warm-up.

100% of about 693 spoofed SYN-flood flows detected in the end-to-end Docker test	0 false alarms on benign HTTP traffic after warm-up calibration	3.8× attack vs benign reconstruction error (0.0365 vs 0.0095)
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Phase 2, in progress: a hybrid setup with a physical Raspberry Pi as Client A alongside Docker clients B and C, an overnight matrix of aggregators across three seeds, and on-device measurements of temperature, power and memory.

07 ROAD TO THE PAPER

- 01 Re-run all aggregators with a matched round budget and at least three seeds, reporting variance.
- 02 Per-attack tables (web vs brute force vs reconnaissance) and TPR at a fixed false-positive rate.
- 03 Model-poisoning experiments against the integrity audit.
- 04 Ablations of local adaptation, blending and the FedProx μ , plus Raspberry Pi edge costs from Phase 2.

Stated limits of the current draft: single-seed numbers, unequal round counts, role-based synthetic partitions of a public dataset, and no physical-edge measurements yet. These are the items above.